AMERICAN INTERNATIONAL UNIVERSITY-BANGLADESH (AIUB)

Course Name: Data Structure Lab

Section: G

Lab Practice Questions: Structures and Arrays

A tech-retail store management system needs to keep track of its active inventory. Define a structure named `Product` containing fields for `productID` (integer), `name` (string), and `price` (float). Write a program that initializes an array of 5 `Product` objects with predefined data, and then implements a linear search function to find and display the full details of a product based on a user-input `productID`. If the ID does not exist in the array, print an appropriate error message.

A university administration office wants to organize student academic records for quick verification. Create a structure called `Student` with members for `studentID` (integer), `gpa` (float), and `creditsCompleted` (integer). Develop a system that stores records for a batch of students in an array, takes a target `studentID` from the user, and applies a linear search algorithm to scan the array. Upon finding the student, the program should output their specific GPA and completed credits; otherwise, it must notify the user that the record was not found.

A city blood bank requires an efficient way to check the availability of specific blood groups during emergencies. Design a structure named `Donor` that holds `donorID` (integer), `bloodGroup` (string), and `age` (integer). Construct an array of `Donor` structures containing records for multiple volunteers, and implement a linear search workflow that prompts an emergency coordinator to enter a required blood group. The program should traverse the array and print the ID and age of all matching donors, or display a message stating no matching donors are available if the group is missing.